Beyond the Black Box:

Interpretability of LLMs in Finance

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Restricted - External

Content

- Why AI interpretability in Finance
- Interpretability Tools – Mechanistic Interp.
- Use Case 1: “The Sentiment Feature”
- Use Case 2: “The Warren Buffet AI”
- Use Case 3: “The Hallucination Police”
- Upcoming work – Agents, Trading and Prediction Models

Paper Link

Why interpretability

85%

AI Pilots don't reach production*. ~70% leaders attribute to explainability **

0%

Financial GenAI tools that looking inside the model

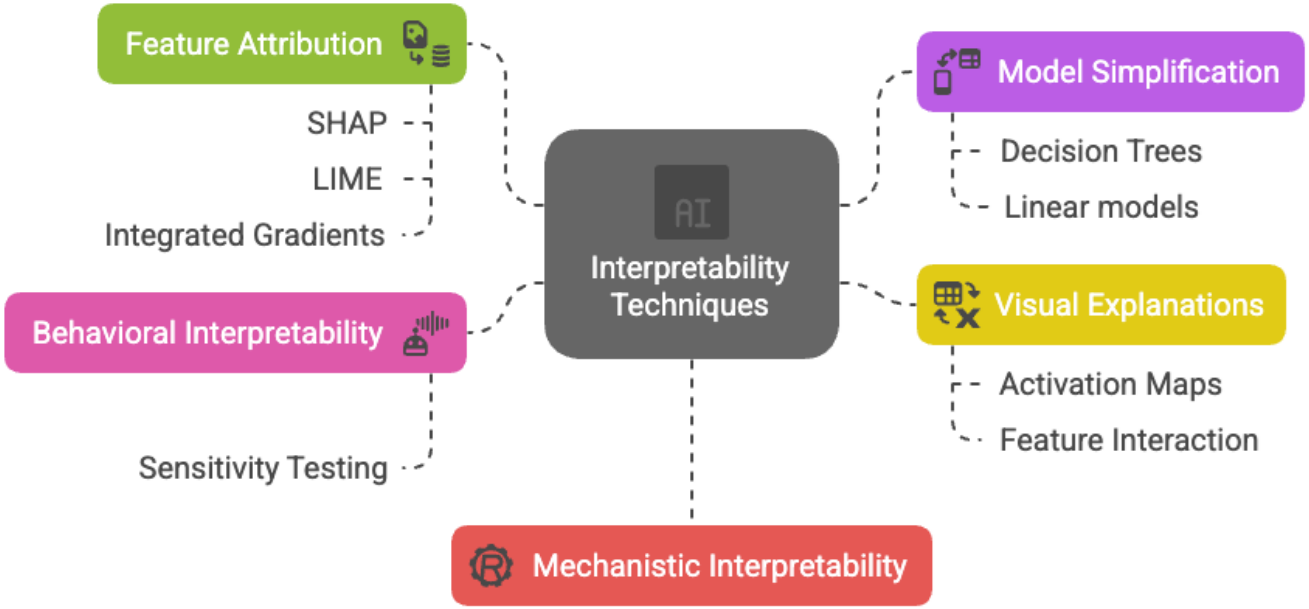
$20T

Cost of not understanding financial models well in 2008.***

* https://www.cio.com/article/3850763/88-of-ai-pilots-fail-to-reach-production-but-thats-not-all-on-it.html
** https://www.ey.com/en_uk/newsroom/2024/12/ey-european-financial-services-ai-survey
***https://www.govinfo.gov/content/pkg/GPO-FCIC/pdf/GPO-FCIC.pdf

Why Interpretability ?

Interpretability Types

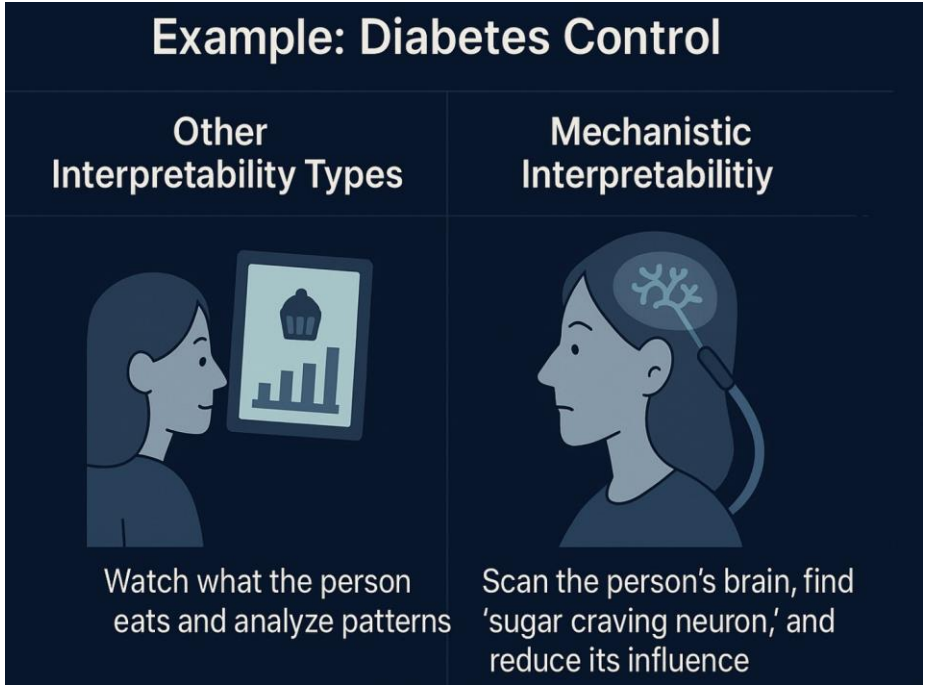


Mechanistic Interpretability

- Neuroscience of Generative AI

- Reverse Engineer internal neurons and layers

- Unlike other types of interpretability techniques – focuses on internal structure



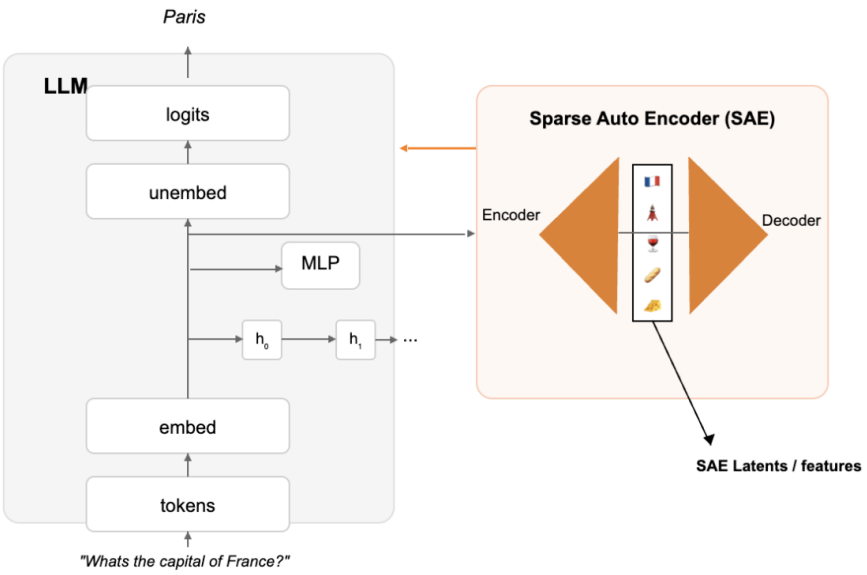
Why Mech. Interp.?

- LLM adoption depends on the transparency.
- Facilitate alignment of Gen AI solutions/models with Model Risk Management and regulatory compliance.
- Finance – stakes related to AI are high, therefore higher model governance requirements

Interpretability Tools

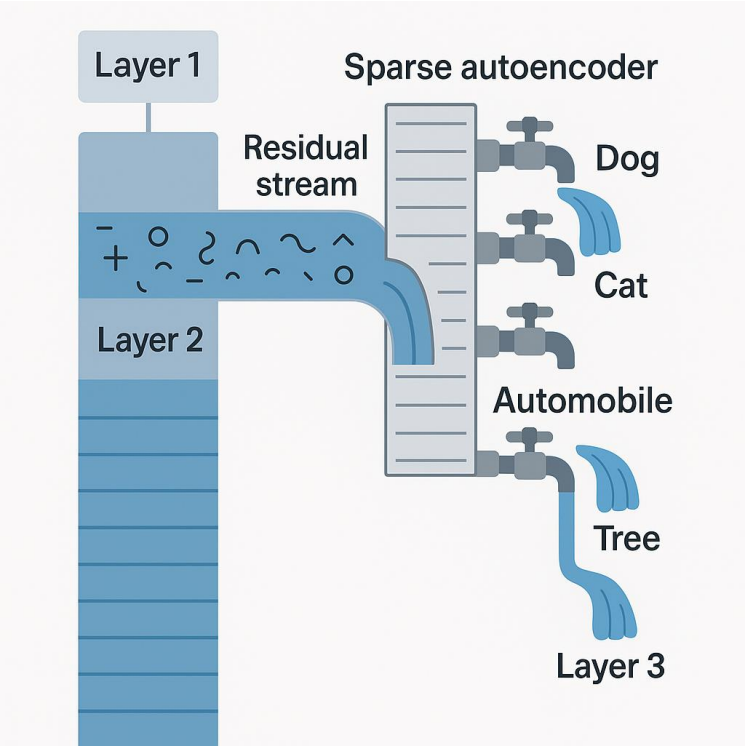
Sparse Autoencoders (SAE)

- Polysematicity – more concepts less #parameters.
- “Astrophysics” and “Michael Jordan” – represented by same neuron
- Zooming into the residual stream using a bigger and sparse Neural network



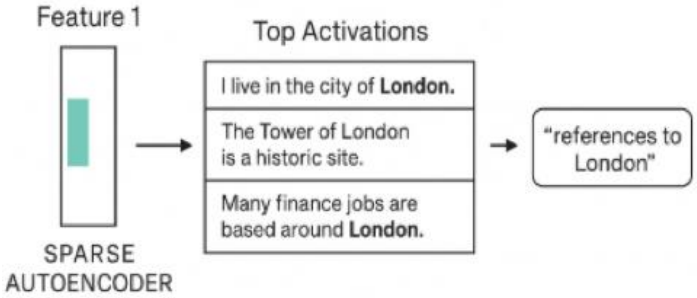
Sparse Autoencoders

- Layer 2 residual stream feeds into SAE.
- SAE decouples separate concepts (dog, cats etc.)

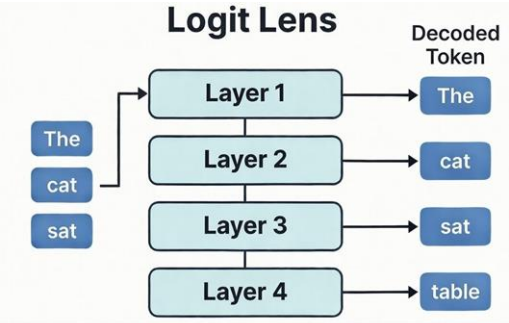


Assigning a human-readable label

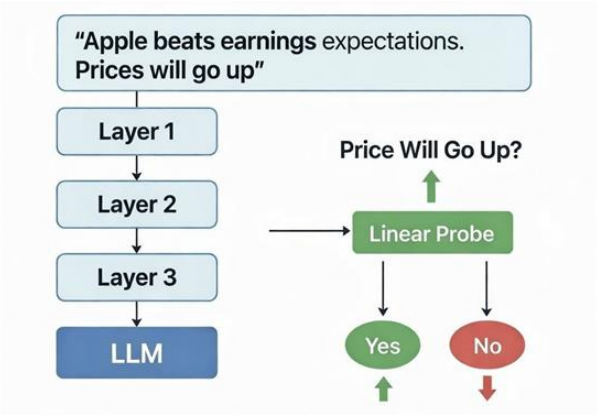
- Labels are assigned using AI by analyzing top text activations
- Example: “references to London” identified from activation patterns



Other Interpretability tools

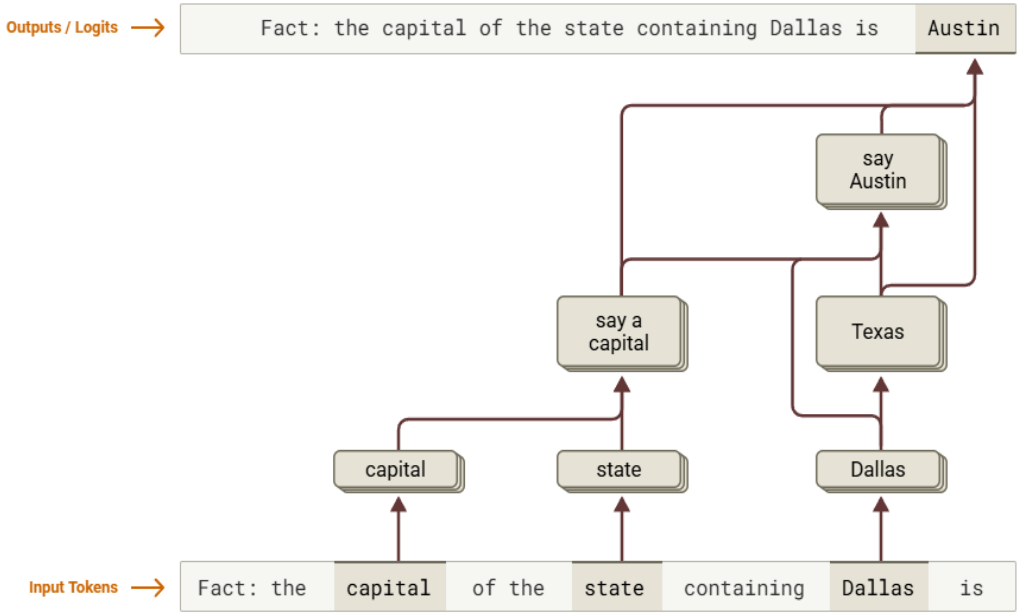


Linear Probe



Other Interpretability tools

Circuits*



*Source : https://transformer-circuits.pub/2025/attribution-graphs/biology.html

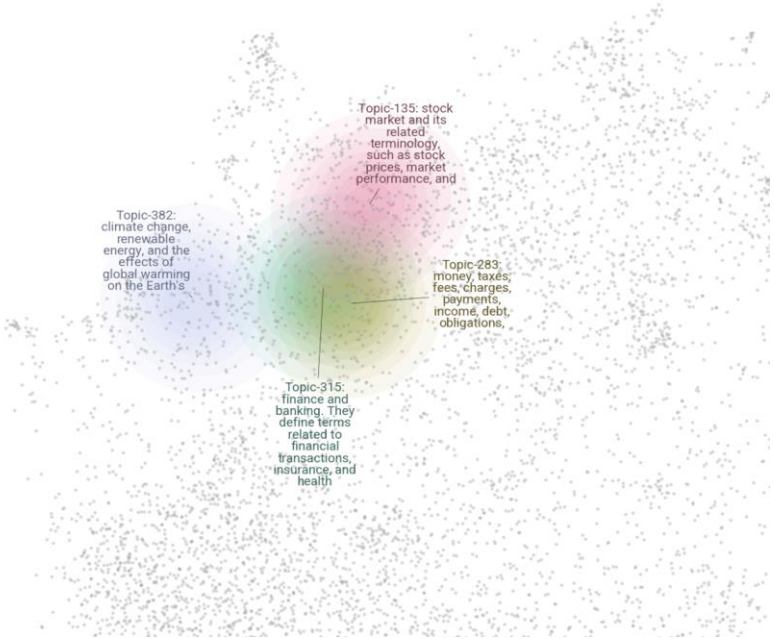
Feature Labelling – Self Interpretability

- Steering - Modify the model’s demeanor, preferences, stated goals, and biases.
- Labeling through steering one feature at a time to explain themselves through self-explain prompt :
    - *What is the meaning of the word "X"?The meaning of the word "X" is*

Restricted - External

Clusters of Finance Features

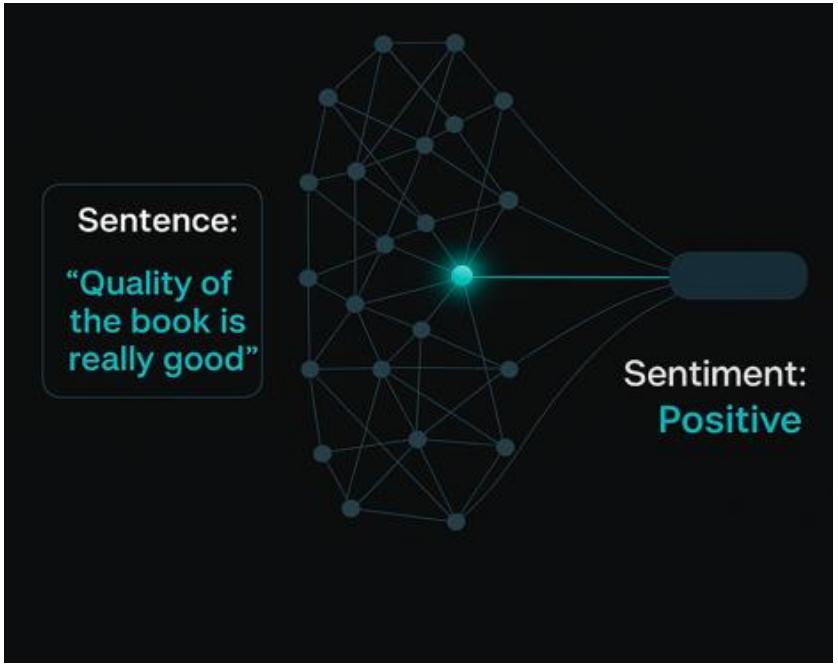
Feature Id	Description
7650	'The concept of tranquility, calmness, and freedom from disturbance or conflict.',
6802	'Geographical boundaries, regions outside, different places, foreign locations, beyond scope.',
8618	'Definitions and examples of terms related to uniqueness and possession.',
2227	'Scientific instruments and devices for observation, measurement, and data collection.',
7254	'Mathematical and numerical concepts related to values, sequences, and magnitude.',
9210	"Google's search engine and related online services using artificial intelligence.",



1.
The Sentiment Feature

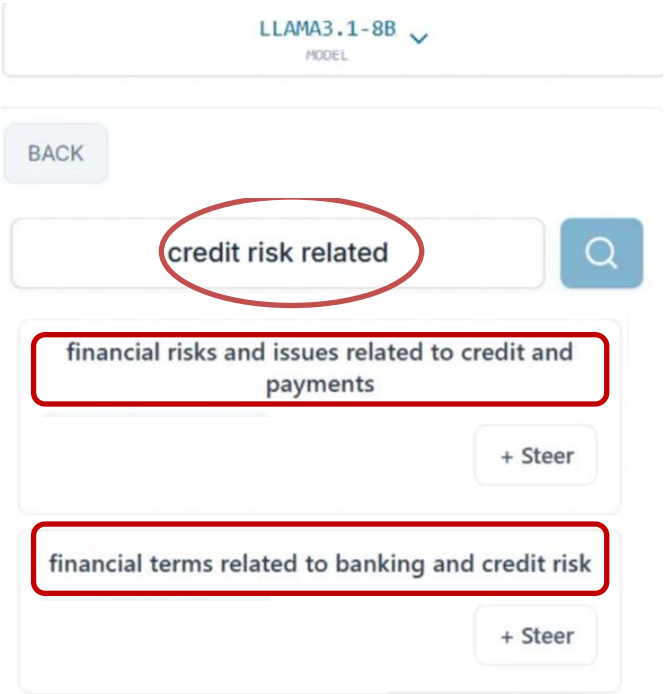
Open AI "Sentiment Neuron" -2017

- Single neuron in an AI model Identified Amazon sentiment.



Source: https://openai.com/index/unsupervised-sentiment-neuron/

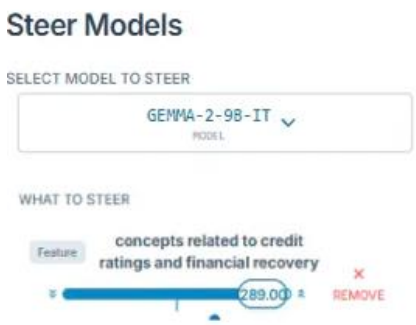
“Credit Risk Sentiment” Features*

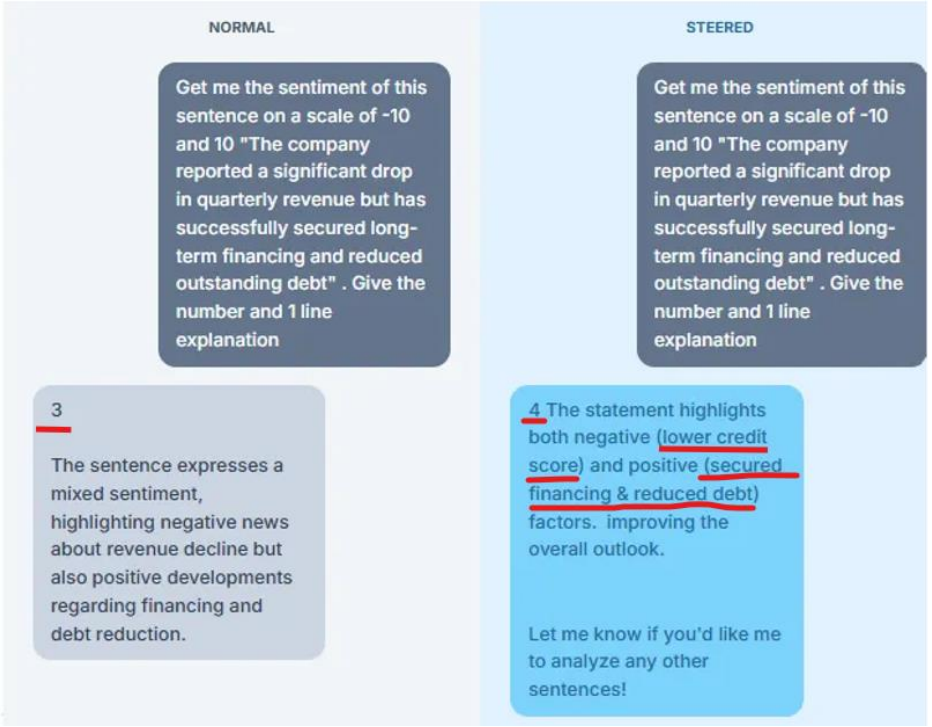


* Source : https://www.neuronpedia.org

Steering Features

- Increase activation of specific feature
- No Prompting
- No finetuning

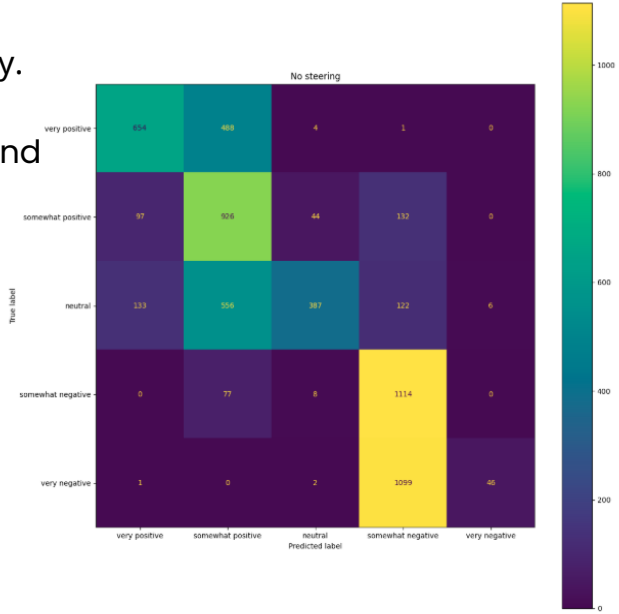


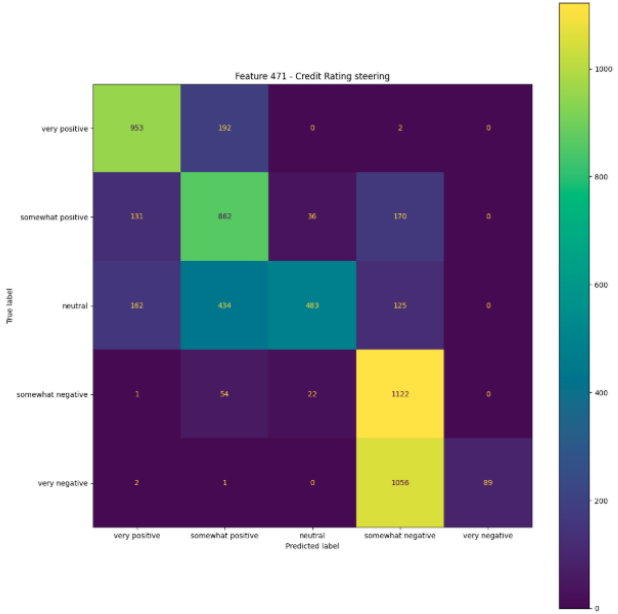


Domain specific “steering”

Impact on Sentiment Analysis – steered for credit risk based feature

- Post steering high accuracy.
- Across Neutral, Positive and negative categories





2.

The Warren Buffet AI

Main argument

“The brain of the LLM is already has all information from the entire internet.”

Including:

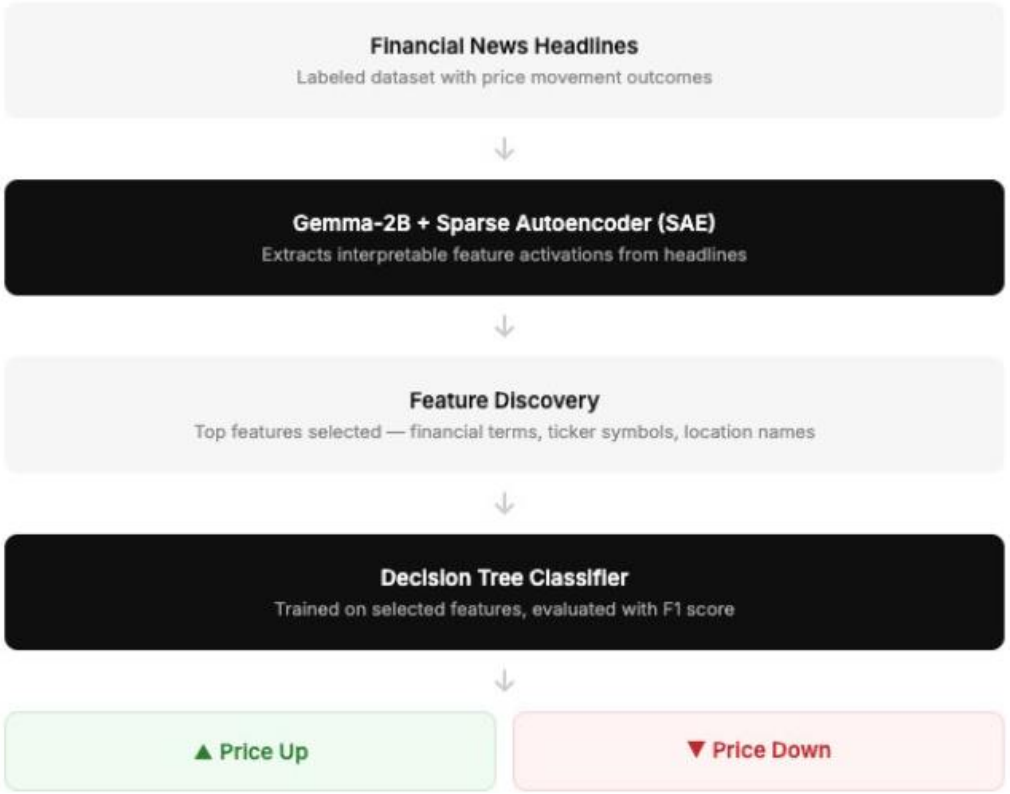
- Pricing anomalies
- Macro regime signals
- Volatility shift patterns
- …even the satellite image information!

Target : Find the zone of ”high” signal-to-noise ratio in the LLM’s brain.

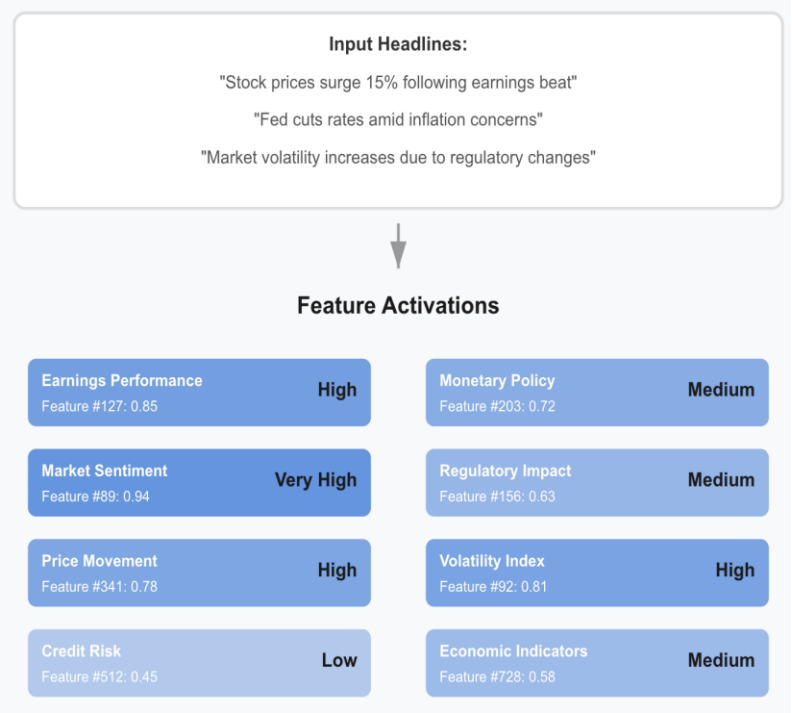
Financial Features and Prediction

- Find “features” that activate on news headlines.
- Train prediction model on selected features

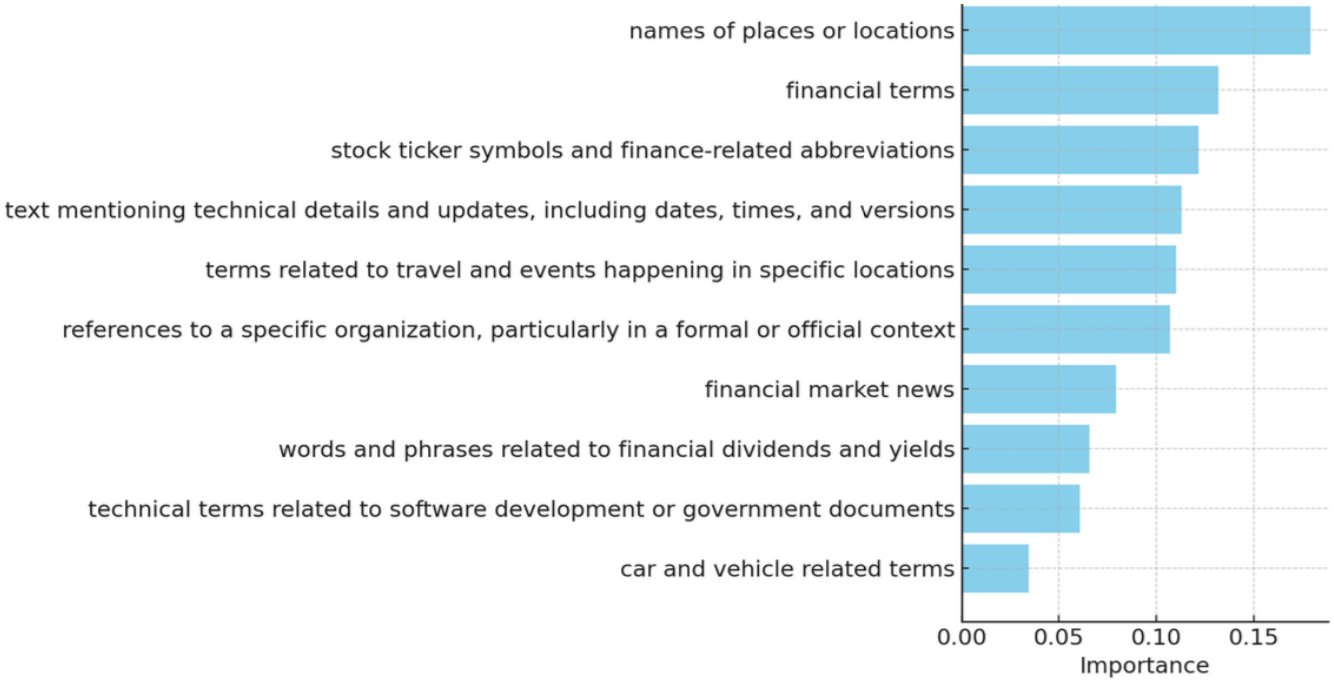
Training Pipeline



Financial Features and activation



Most relevant features



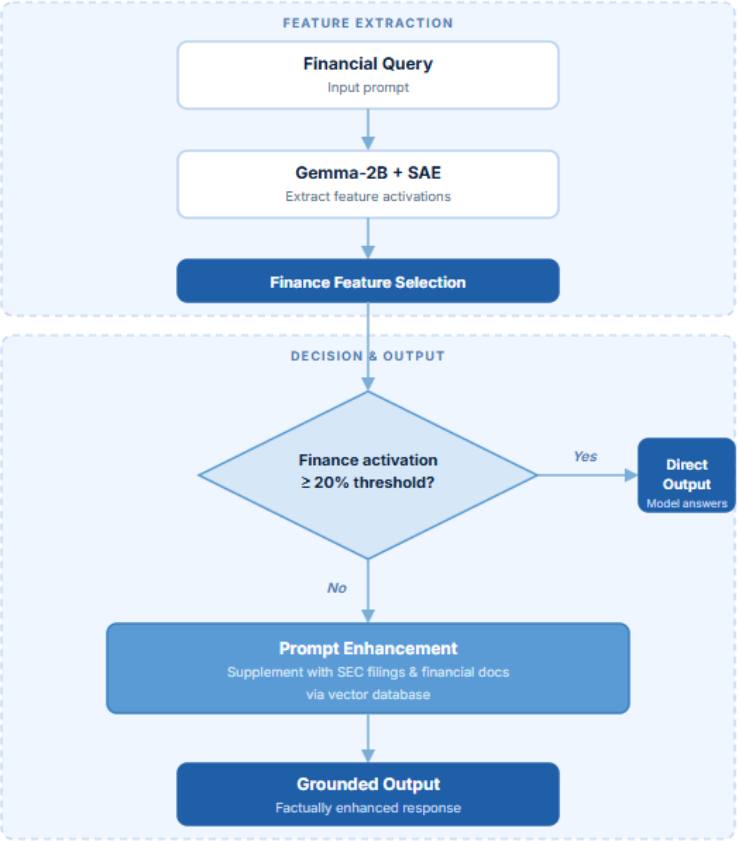
3.

The “Hallucination” Police

Hallucination reduction

- Select finance-specific features.
- Monitor activation percentages.
- Below threshold triggers prompt enhancement.

Steps


FEATURE EXTRACTION
Financial Query
Input prompt
Gemma-2B + SAE
Extract feature activations
Finance Feature Selection
DECISION & OUTPUT
Finance activation
≥ 20% threshold?
Yes
Direct Output
Model answers
No
Prompt Enhancement
Supplement with SEC filings & financial docs
via vector database
Grounded Output
Factually enhanced response

Sample Features

Feature	Mean Activation	Feature Label
15857	10.842405	**phrases related to financial terms and information**
5405	7.893589	phrases related to market statistics and analysis
11912	2.604609	text mentioning technical details and updates, including dates, times, and versions
6513	2.041639	references to education and business institutions, particularly relating to specific programs, schools, and disciplines
14291	1.862908	financial news related to the stock market
7906	1.781035	mentions regarding customer interaction and feedback in a business context
15354	1.751971	**terms related to financial news and companies**
1213	1.395549	dialogue and text indicating interactions among users in an online forum setting
8312	1.293682	technical terms and administrative terms related to website administration and online forums
3553	1.130796	names of places or locations.
544	1.129925	terms related to medical conditions and medications, specifically focusing on cholesterol-lowering drugs
354	1.013154	words related to financial markets, funds, and investments
6510	0.936306	character sequences indicating the start of new sections or categories
15173	0.901733	terms related to real estate and local government activities
3586	0.857892	attends from the beginning-of-sequence marker to the first token in the sentence.

Hallucination reduction (2 /2)

Results

Actual Prompt – No conditional check	Enhanced prompt – With conditional check
Companies must file Form 8-K to report significant corporate events in a timely manner, typically within four business days of the event. This keeps investors informed about material changes.	**Activation is below threshold.** According to SEC regulations, Form 8-K must be filed within four business days after the occurrence of a triggering event for most items. However, Item 7.01 (Regulation FD Disclosure) requires filing on the same day. Item 4.02 (Non-Reliance on Previously Issued Financial Statements) has specific timing requirements based on when the determination is made. Late filings can result in SEC enforcement actions and loss of Form S-3 eligibility. Companies must maintain disclosure controls to ensure timely 8-K filings. Electronic filing through the EDGAR system is mandatory. Source: SEC Final Rule 33-8400 (17 CFR Parts 228, 229, 230, 239, 240 and 249)

Upcoming Work

Augmenting classical factors

- Discover unstructured data factors, joined with classical factor structure
- Integrate with classic statistical techniques
- Leverage for Trading and factor modeling.

Classical Factor
S&P 500 Return
Industrial Sector ETF Return
Daily Change in VIX
20-Day Momentum
20-Day Rolling Volatility
Volatility Z-Score

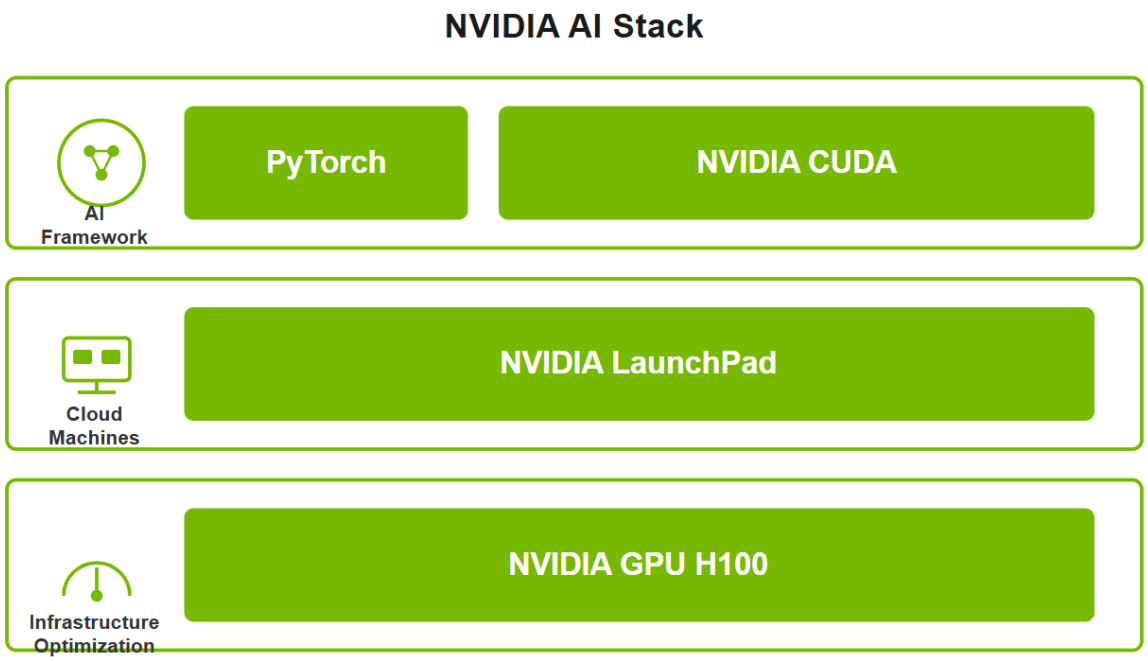
LLM Internals (SAE Features)

Feature ID	Feature Name
79	Insider buying; large insider share purchases
257	Earnings misses relative to estimates
319	Ellipsis/uncertainty markers in text ("...")
100	Analyst blog/commentary on stock performance
149	M&A / deal amount and contract-value language
100	Analyst blog/commentary on stock performance

Agentic Monitoring

- Monitor agent reasoning **during execution** using internal signals.
- Use **internal feature** activations as a live diagnostic signal
- Detect **tool-call misfires** or off-path reasoning in real time
- Intervene by pausing or adjusting the agent before continuation
- Improves reliability across agent workflows

NVIDIA Stack



Current Limitations

- Feasible for **open-weight models** → Encourage vendors to have more open-weights models
- Closed-weight model vendors should provide interpretability tools
- Might be limited to few layers of the LLM, circuit tracing could unlock multi-layer behavior
- Limited evaluation metrics of interpretability, ongoing research

Final Thoughts

AI interpretability is a must in Finance and high-stakes domains - it:

- Unlocks new use cases, scales adoption across our industry
- Builds regulator trust, allows for wider deployment
- May avoid a financial crisis caused by not understanding the model well.

Thank You!

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